

Small store food purchase behavior

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2026-04-04

The STORE survey

- In the Fall of 2014, the City of Minneapolis revised an ordinance requiring all stores with grocery licenses to carry a wide array of healthy foods and beverages in order to better align with the Dietary Guidelines for Americans and stocking requirements for WIC (Women, Infants and Children) program vendors.
- To date, Minneapolis is the only city in the U.S. to have a policy of this kind in place. This type of policy action may have an important impact on food access, particularly in under-served neighborhoods and small food stores that do not currently stock these types of items.
- The purpose of this **STaple foods ORdinance Evaluation (STORE) study**, therefore, is to evaluate the impact of this local policy change (i.e., the Minneapolis Staple Foods Ordinance) that establishes minimum stocking criteria for a wide array of healthy foods as a requirement of food store licensing.
- One of the type of data collected were about the nutritional quality of consumer purchases at small-to mid-sized food stores, including assessment of energy density and calories via customer intercept surveys and direct observation of purchases.
- The stores included in the study were: convenience store, food-gas stores, dollar stores and pharmacies.
- The data set `small_store_customer_intercept.csv`, includes information about the purchase amount and the diet quality of food item purchased by 2760 customers of small and medium food stores, in Minneapolis and St. Paul, Minnesota.
- Many demographic variables (`age_group`, `education`, `sex`, `BMI_group`) can help us understand the customer characteristics of small and medium stores that carry food items.
- Recorded Healthy Eating Index data:
 - The variable `hei2010_total_score` records the Healthy Eating Index-2010 (HEI-2010) for each customer that evaluates the quality of Diet in a 100-point scale.
 - The `hei_group` variable categorize each customer to a High, Good, Average, Needs Improvement and Poor level of quality of diet based on the interpretation of their `hei2010_total_score`.
 - The `hei_4groups` splits the `hei2010_total_score` sample data to 4 equal groups and allows us to see the distribution of the scores in higher resolution.
- Other interesting variables are :
 - `shopfreq3`: the frequency that customers in Minnesota purchase food items from small and medium sized stores
 - `addsugar`: the percent of calories from added sugars (among food/beverages purchased)

- rikcal: The energy (kilocalories) of food/beverages purchased
 - fruit_tot, veg_tot, whgrains_all_tot, lf_milk_tot: indicator variables that record if the purchase included at least one serving of fruit, vegetables, whole grains or low fat milk correspondingly.
- Analyzing the above variables we can:
 - get an estimate of the diet quality of the population in Minneapolis and St. Paul.
 - understand the typical customer of small and medium food stores.
 - find the groups that follow the poorest diet quality based on their purchases from small and medium food stores.
 - Use the above information to inform policies that increase the quality of eating habits.

Load the customer-diet quality data

```
library(tidyverse)
library(ggmosaic)

customer_intercep <- readRDS("small_store_customer_intercept.csv")

head(customer_intercep)
```

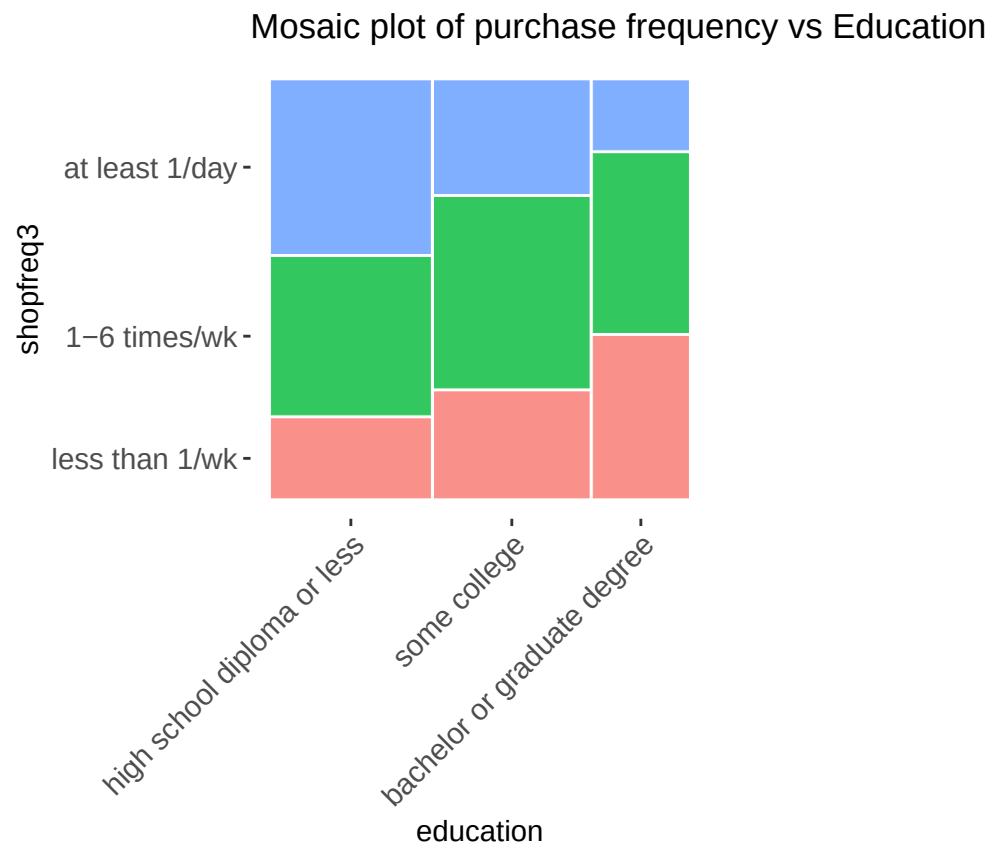
```
##   storeid time      stcity items_count age hei2010_total_score purchase_cost
## 1    1001   1 Minneapolis         1  25          20.00000         6.13
## 2    1001   2 Minneapolis         5  24          36.28802         8.07
## 3    1001   3 Minneapolis         1  26          11.72840         3.00
## 4    1001   4 Minneapolis         1  65          30.00000         0.50
## 5    1001   1 Minneapolis         1  21          20.00000         1.86
## 6    1001   2 Minneapolis         2  22          20.00000         1.38
##      rikcal employment3      education race_new      shopfreq3 male
## 1 1635.696    employed      some college      1 1-6 times/wk    1
## 2 2683.344  unemployed bachelor or graduate degree      1 1-6 times/wk    0
## 3  166.050    employed      some college      1 at least 1/day    1
## 4  124.030      other high school diploma or less      1 at least 1/day    1
## 5  246.400    employed      some college      1 1-6 times/wk    1
## 6  272.616      other      some college      1 1-6 times/wk    1
##   obese overweight fruit_tot veg_tot whgrains_all_tot lf_milk_tot      satfat
## 1     0           0         0         0              0          0 0.0000000
## 2     0           0         1         0              0          0 0.5218861
## 3     0           0         0         0              0          0 0.0000000
## 4     0           0         0         0              0          0 0.1451262
## 5     0           1         0         0              0          0 0.0000000
## 6     0           0         0         0              0          0 0.0000000
##   addsugar      sodium  density      sex  age_group BMI_group      hei_4groups
## 1 96.97303 108.10811      NA    male (17.9,26.9]    normal (-0.0888,22.2]
## 2 19.37165 4794.33013 1.225313 female (17.9,26.9]    normal  (22.2,44.4]
## 3 89.42126 1844.44444      NA    male (17.9,26.9]    normal (-0.0888,22.2]
## 4 69.16391  88.60759 0.790000 male (62.4,71.2]    normal  (22.2,44.4]
## 5 89.80032 225.00000      NA    male (17.9,26.9] overweight (-0.0888,22.2]
## 6 96.97303 108.10811      NA    male (17.9,26.9]    normal (-0.0888,22.2]
##   hei_group
```

```
## 1    Poor
## 2    Poor
## 3    Poor
## 4    Poor
## 5    Poor
## 6    Poor
```

The demographic characteristics of the most frequent customers of small and medium food stores.

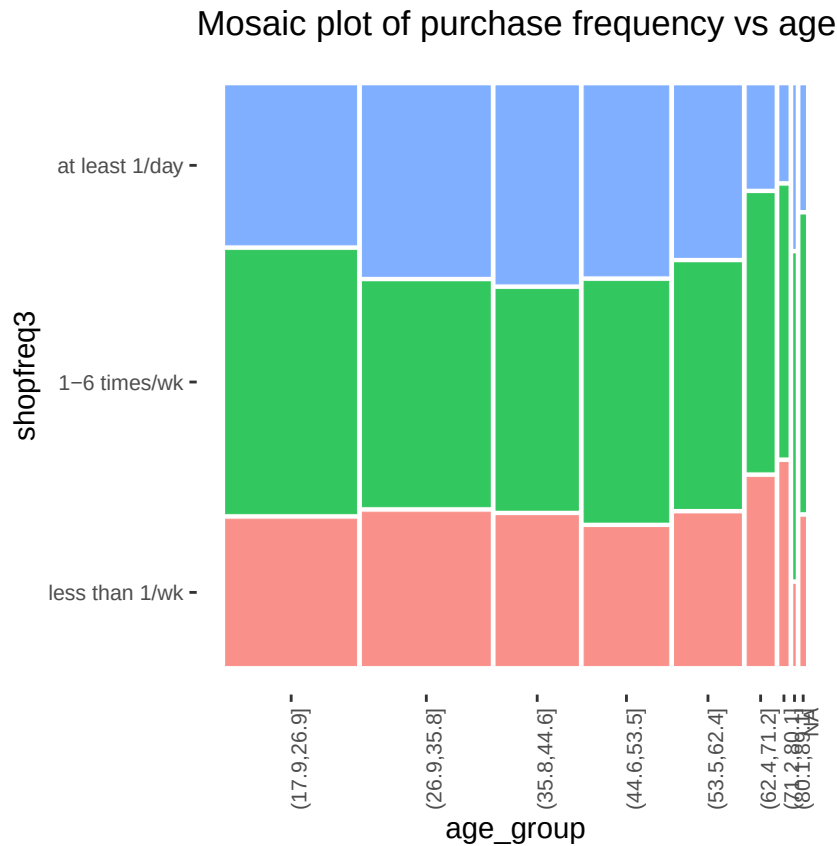
- High school diploma or less level of education has the highest proportion of customers that buy products from a small/medium store at least once per day.

```
ggplot(customer_intercep) +
  geom_mosaic(aes(x = product(shopfreq3, education),
                    fill=shopfreq3),
              show.legend = FALSE) +
  theme_mosaic() +
  theme(axis.text.x = element_text(angle = 45,
                                    hjust = 1, vjust = 1),
        axis.text = element_text(size = 11)) +
  ggtitle("Mosaic plot of purchase frequency vs Education")
```



- The 26 to 53.5 years old age groups have the highest proportion of customers that buy products from a small/medium store at least once per day.

```
ggplot(customer_intercep) +
  geom_mosaic(aes(x = product(shopfreq3, age_group),
                  fill=shopfreq3),
              show.legend = FALSE) +
  theme_mosaic() +
  theme(axis.text.x = element_text(angle = 90,
                                    hjust = 1, vjust = 1),
        axis.text = element_text(size = 8)) +
  ggtitle("Mosaic plot of purchase frequency vs age")
```



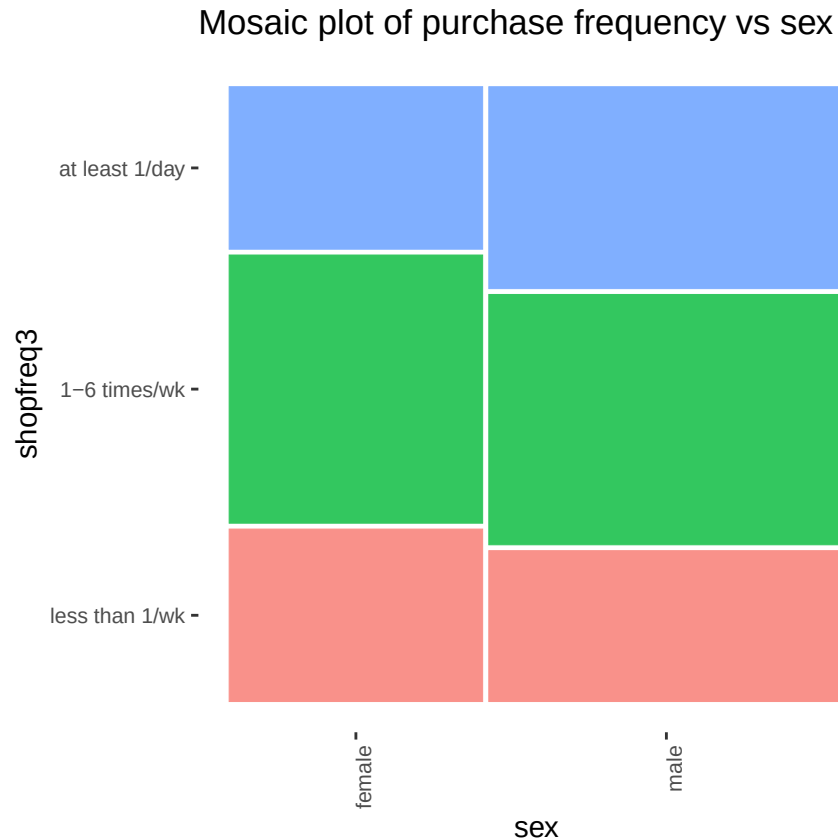
- About the same proportions of female and male customers visit small/medium food stores in all frequency levels but the proportion of male customers that shop at least once per day is higher than the female proportion of our sample.
- A chi square test of independence can validate that the shopping frequency distribution is different between male and female customers.
- The code below includes the test.

```
ggplot(customer_intercep) +
  geom_mosaic(aes(x = product(shopfreq3, sex),
                        fill=shopfreq3),
```

```

    show.legend = FALSE) +
  theme_mosaic() +
  theme(axis.text.x = element_text(angle = 90,
                                    hjust = 1, vjust = 1),
        axis.text = element_text(size = 8)) +
  ggtitle("Mosaic plot of purchase frequency vs sex")

```



```

twoway_table <- table(customer_intercep$shopfreq3, customer_intercep$sex)
chisq.test(twoway_table)

```

```

##
## Pearson's Chi-squared test
##
## data: twoway_table
## X-squared = 13.912, df = 2, p-value = 0.000953

```

Exploring the HEI index within different demographic categories

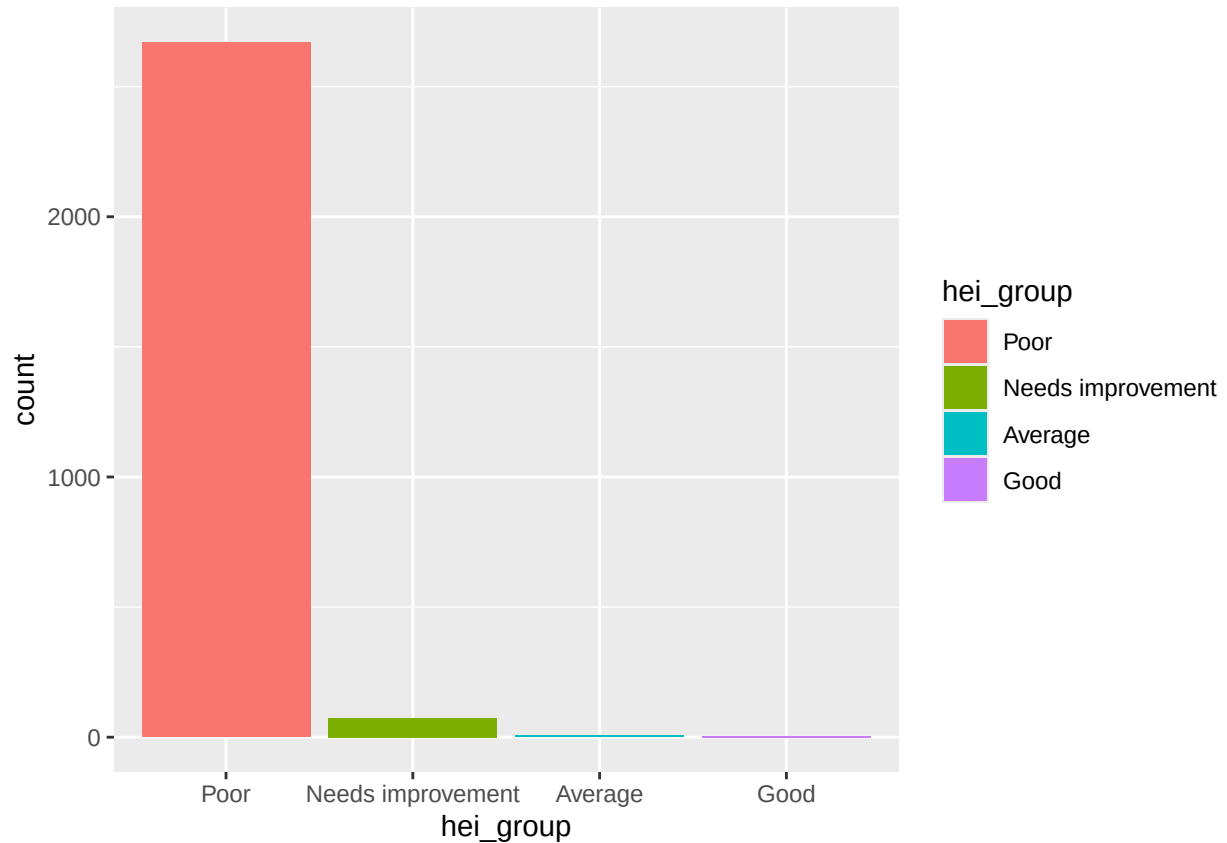
Diet quality groups in Minnesota

- The HEI score is interpreted as follows:
 - (90–100): High-quality diet.

- (80–89): Good quality diet.
- (70–79): Average diet.
- (60–69): Needs improvement.
- (0–59): Poor diet quality.

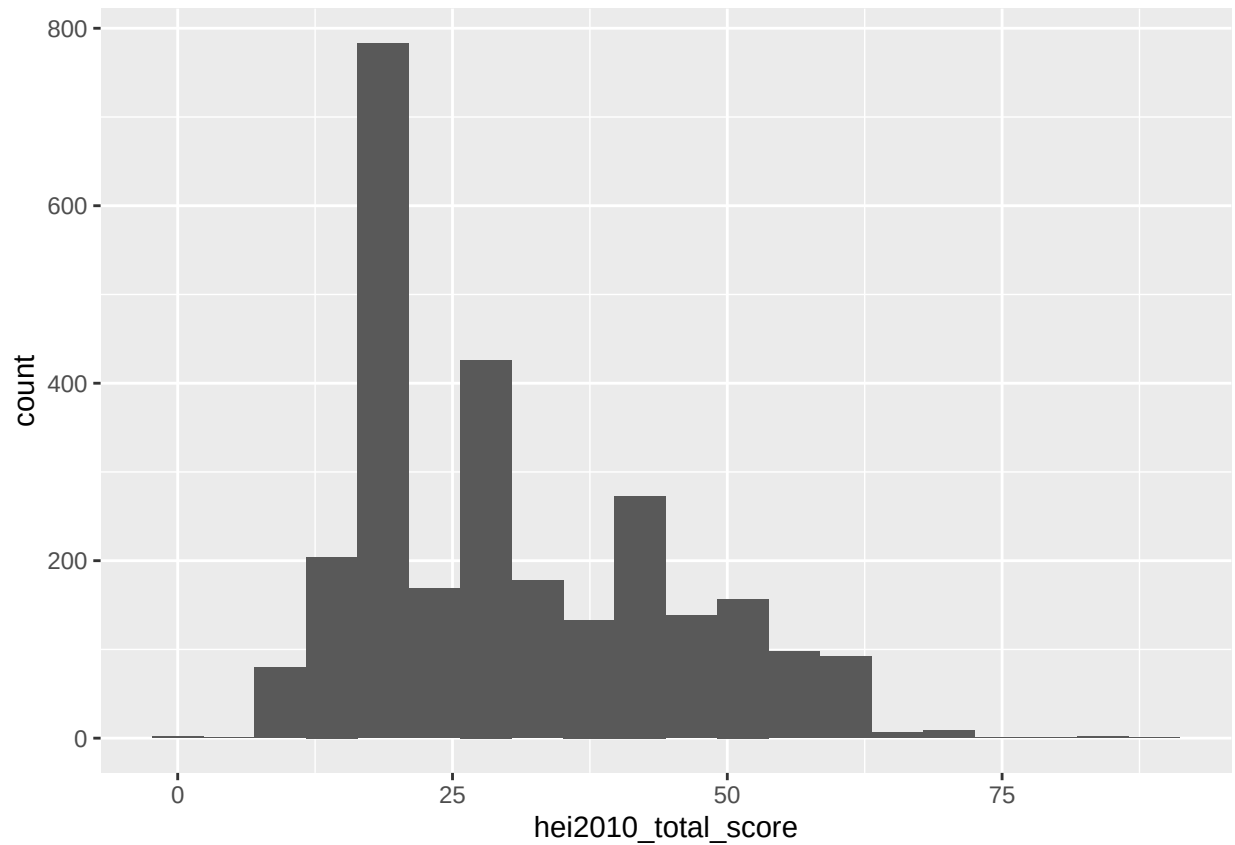
- Based on the sample data the majority of customers purchased food that indicates poor diet quality.

```
ggplot(customer_intercep, aes(x=hei_group, fill = hei_group)) +  
  geom_bar()
```



* There are customers with negative score and unfortunately only a few with score above 80.

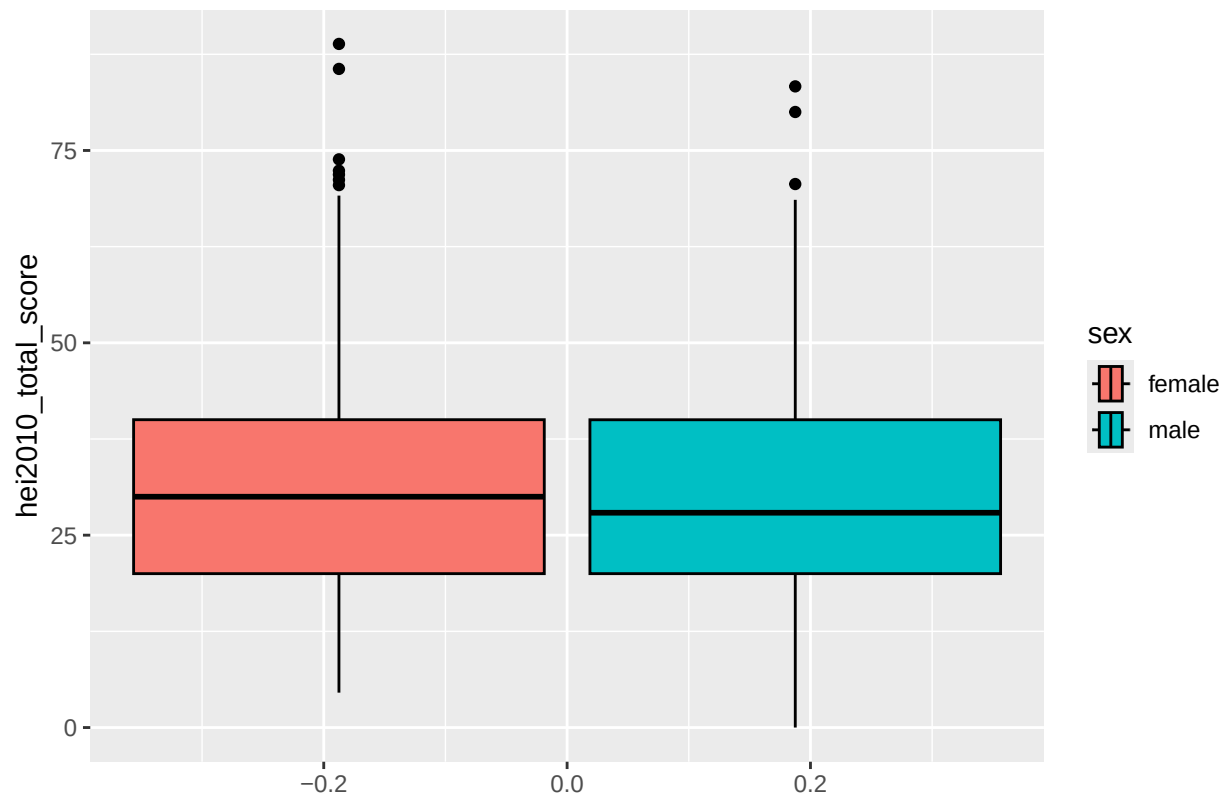
```
ggplot(customer_intercep, aes(x=hei2010_total_score)) +  
  geom_histogram(bins = 20)
```



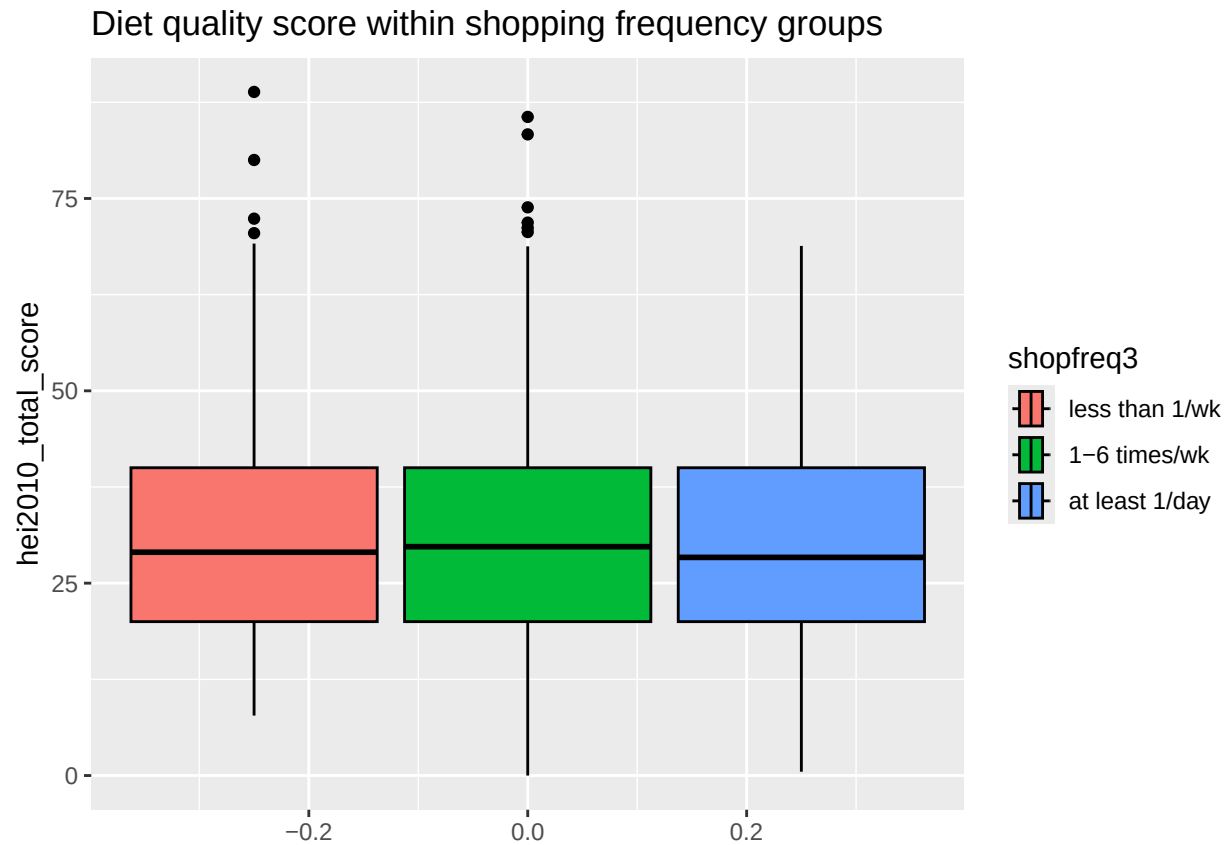
The HEI scores seem different within the two sex groups

```
ggplot(customer_intercep, aes(x=hei2010_total_score,  
                             fill = sex)) +  
  geom_boxplot( color = "black") +  
  coord_flip()+  
  ggtitle("Diet quality score within sex groups")
```

Diet quality score within sex groups



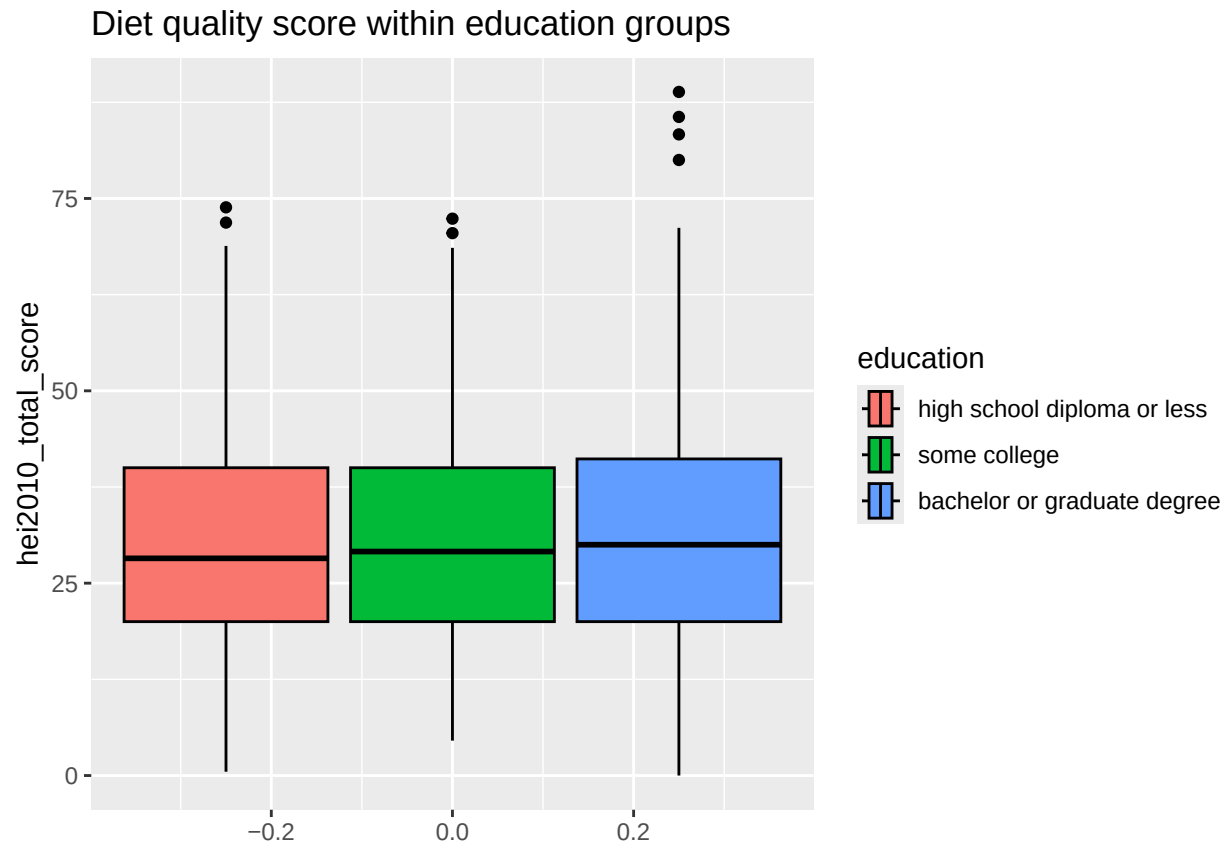
```
ggplot(customer_intercep, aes(x=hei2010_total_score,  
                             fill = shopfreq3)) +  
  geom_boxplot( color = "black") +  
  coord_flip()+  
  ggtitle("Diet quality score within shopping frequency groups")
```



The HEI scores seem different within the different education groups

- College graduates seem to have a better average HEI score.

```
ggplot(customer_intercep, aes(x=hei2010_total_score,  
                             fill = education)) +  
  geom_boxplot( color = "black") +  
  coord_flip()+  
  ggtitle("Diet quality score within education groups")
```

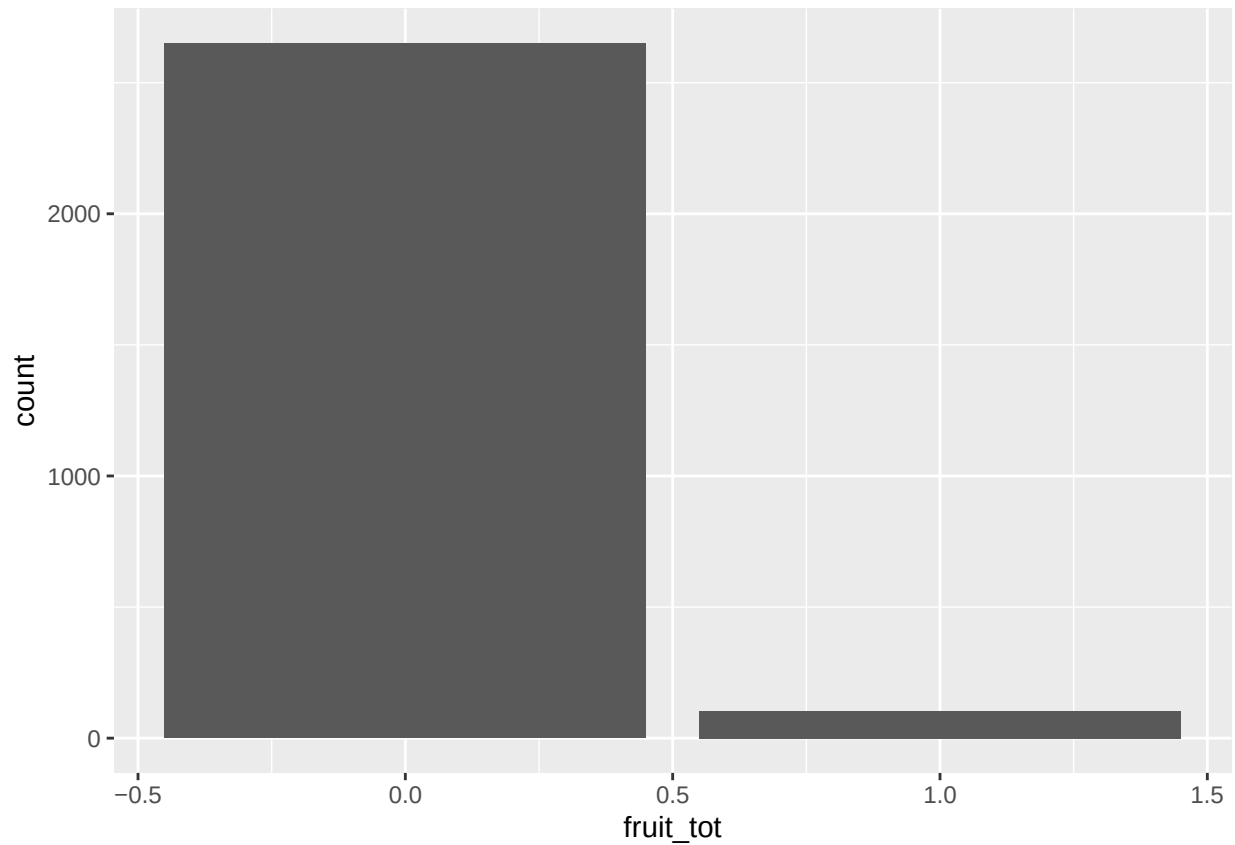


Healthy objects purchased

- Very few customers include at least one serving of fruits or vegetables or whole grains.

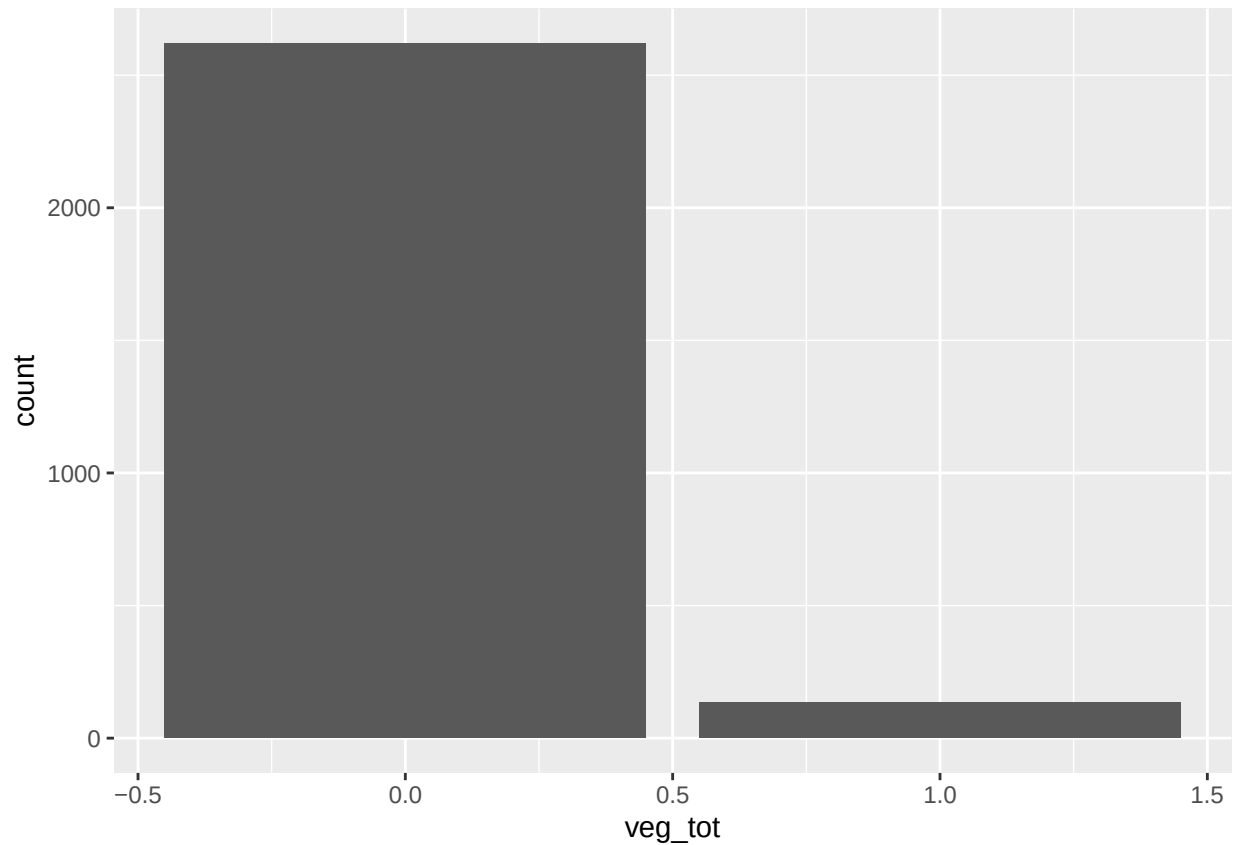
```
ggplot(customer_intercep, aes(x=fruit_tot, fill = fruit_tot)) +  
  geom_bar()
```

```
## Warning: The following aesthetics were dropped during statistical transformation: fill.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
##   the data.  
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
##   variable into a factor?
```



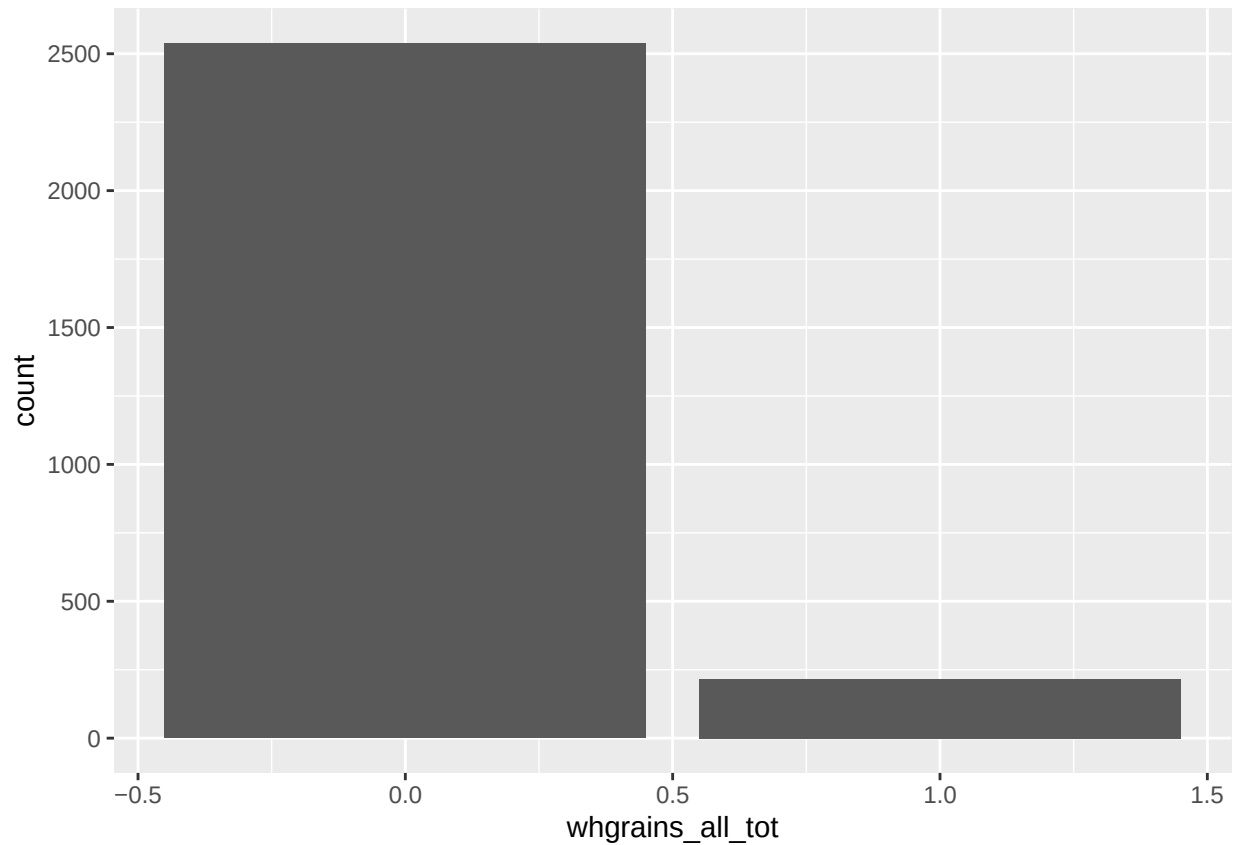
```
ggplot(customer_intercep, aes(x=veg_tot, fill = veg_tot)) +  
  geom_bar()
```

```
## Warning: The following aesthetics were dropped during statistical transformation: fill.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
##   the data.  
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
##   variable into a factor?
```



```
ggplot(customer_intercep, aes(x=whgrains_all_tot, fill = whgrains_all_tot)) +  
  geom_bar()
```

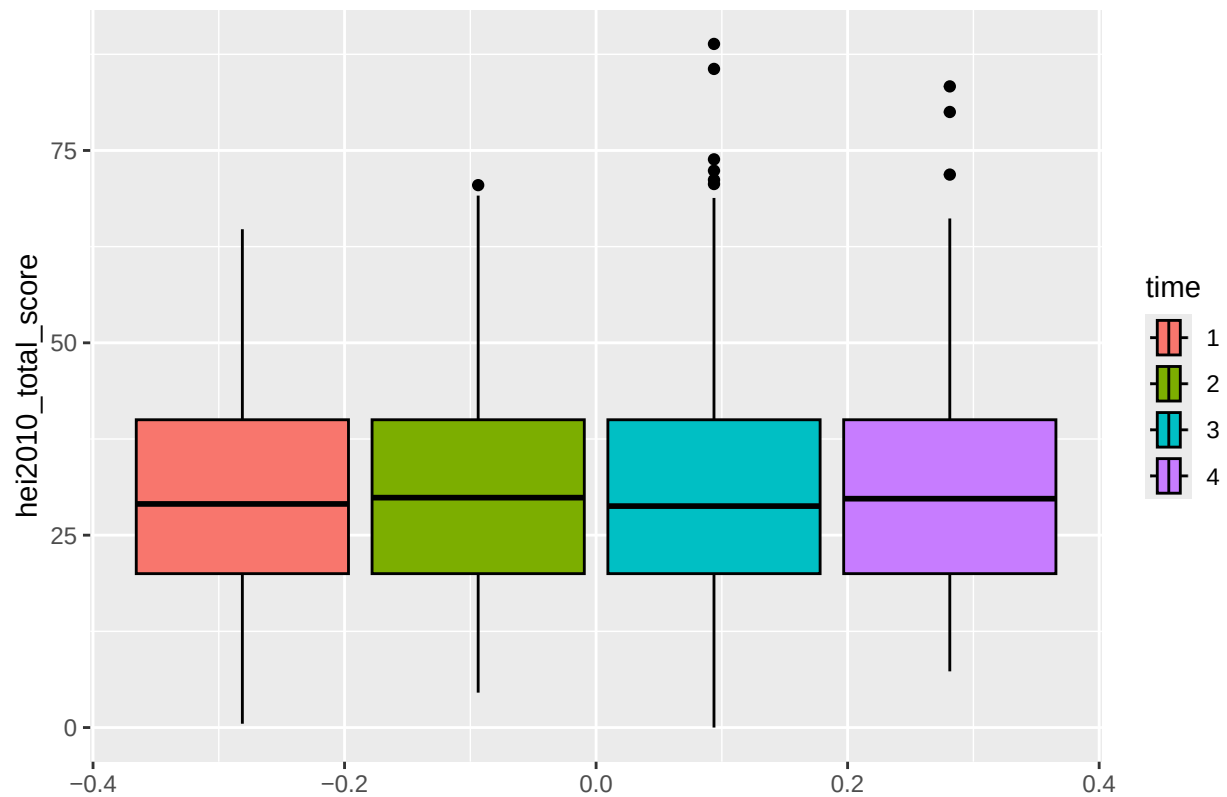
```
## Warning: The following aesthetics were dropped during statistical transformation: fill.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
##   the data.  
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
##   variable into a factor?
```



Did the HEI score improve after the application of the new measures (time = 2,3,4)?

```
ggplot(customer_intercep, aes(x=hei2010_total_score,  
                             fill = time)) +  
  geom_boxplot( color = "black") +  
  coord_flip()+  
  ggtitle("Diet quality score within time groups")
```

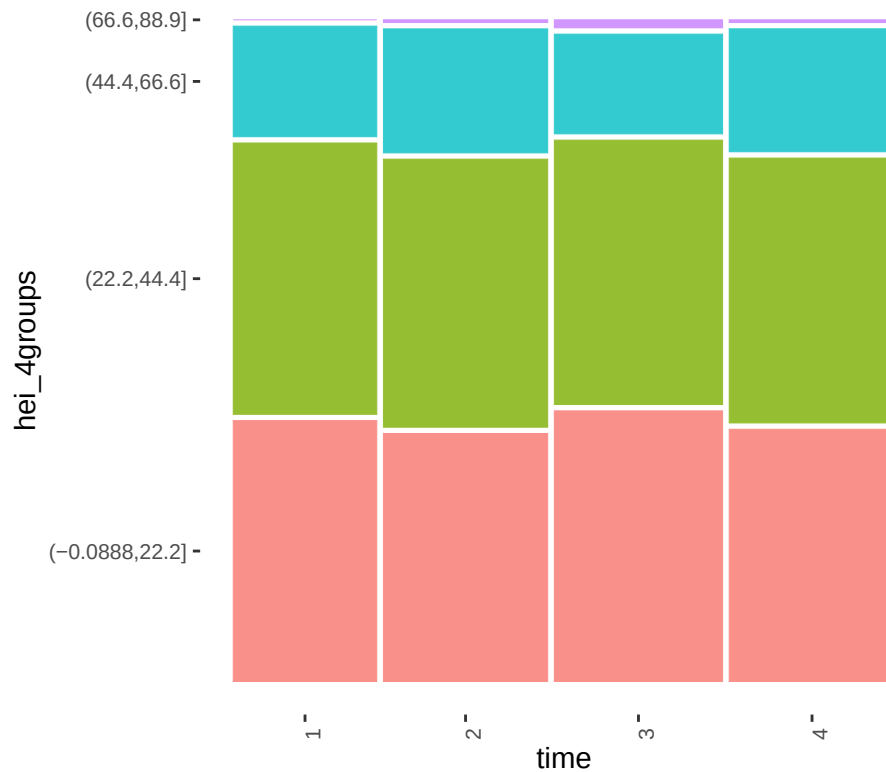
Diet quality score within time groups



- A chisquare test can tell us if the distribution of HEI score groups has changed after the 1st time period. The mosaic plot shows small shifts.

```
ggplot(customer_intercep) +
  geom_mosaic(aes(x = product(hei_4groups, time),
                    fill=hei_4groups),
              show.legend = FALSE) +
  theme_mosaic() +
  theme(axis.text.x = element_text(angle = 90,
                                    hjust = 1, vjust = 1),
        axis.text = element_text(size = 8)) +
  ggtitle("Mosaic plot of hei group vs time period after measure applied")
```

Mosaic plot of hei group vs time period after measure appl



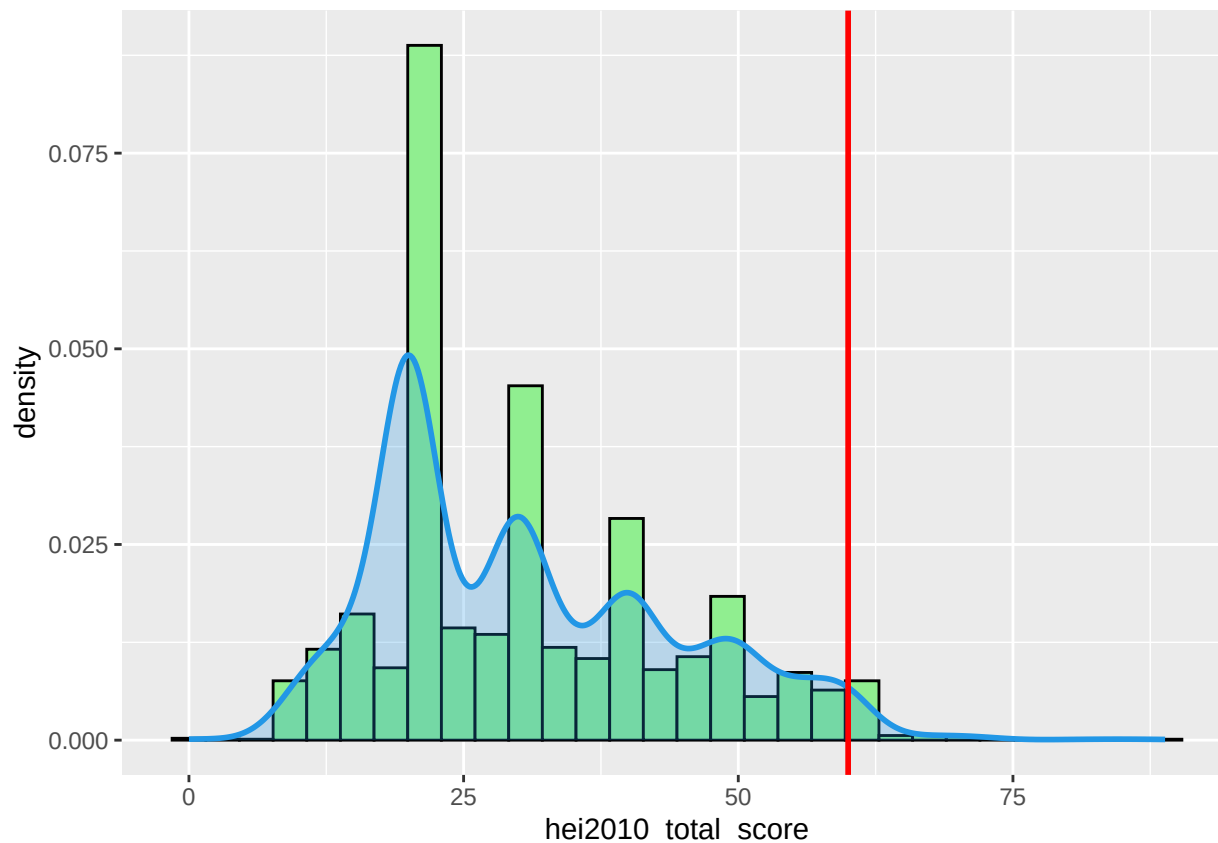
Questions/tasks to explore:

1. Create the histogram of HEI score (hei2010_total_score) and overlay the estimated density. Example code

```
ggplot(customer_intercep, aes(x = hei2010_total_score)) +
  geom_histogram(aes(y = ..density..),
    colour = 1, fill = "lightgreen") +
  geom_density(lwd = 1, colour = 4,
    fill = 4, alpha = 0.25) +
  geom_vline(xintercept = 60, color = "red", linewidth = 1)
```

```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(density)' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



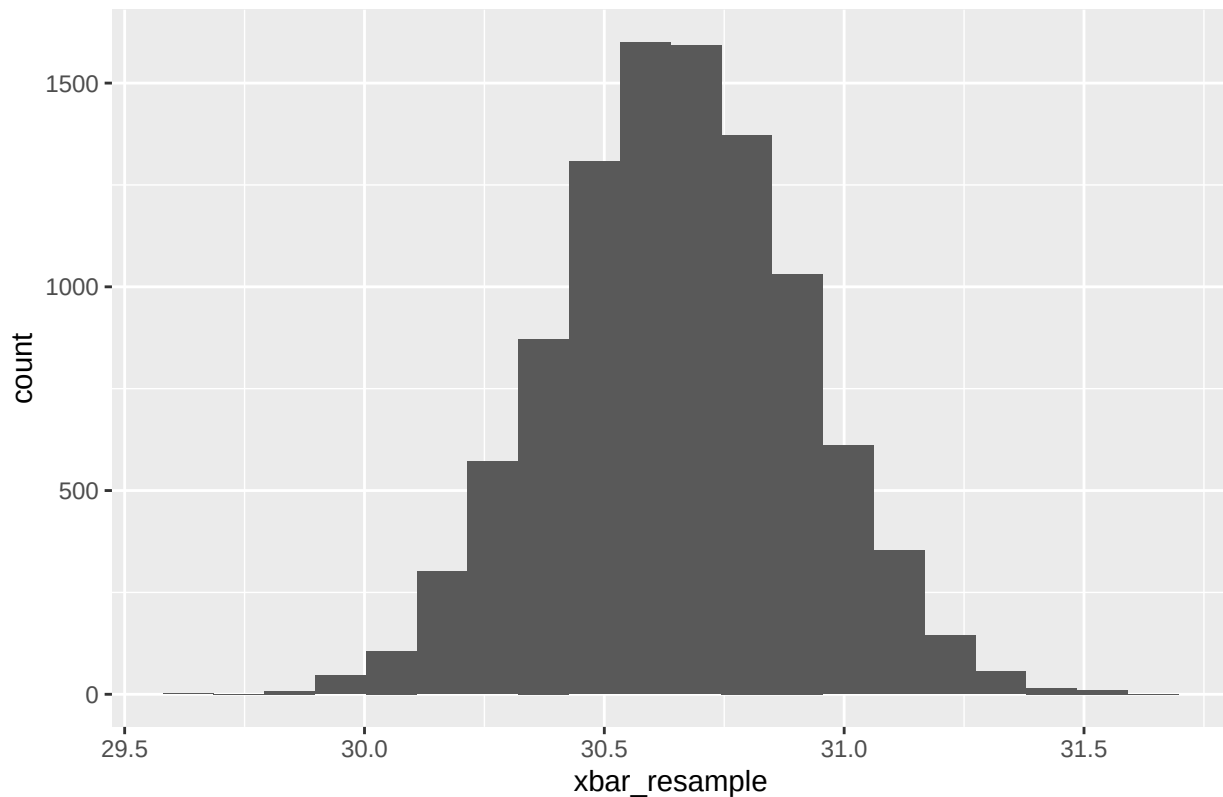
The distribution of HEI 2010 total scores appears right-skewed, indicating that most individuals have moderate diet quality, with fewer individuals achieving very high scores. The peak of the distribution occurs around 24, suggesting this is the most common score range. The spread of the data shows that HEI scores vary across individuals, ranging approximately from 7 to 70. The density curve reinforces this pattern, highlighting the overall shape and skewness of the distribution.

2. Estimate the mean HEI score (`hei2010_total_score`) of customers in Minnesota. Bootstrap CI for a mean - Lecture 23.

```
xbar_resample <- replicate(10000, {
  customer_resample <- sample(customer_intercep$hei2010_total_score, replace = TRUE)
  mean(customer_resample)
})
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  ggtitle("Estimated sampling distribution of xbar")
```

Estimated sampling distribution of xbar

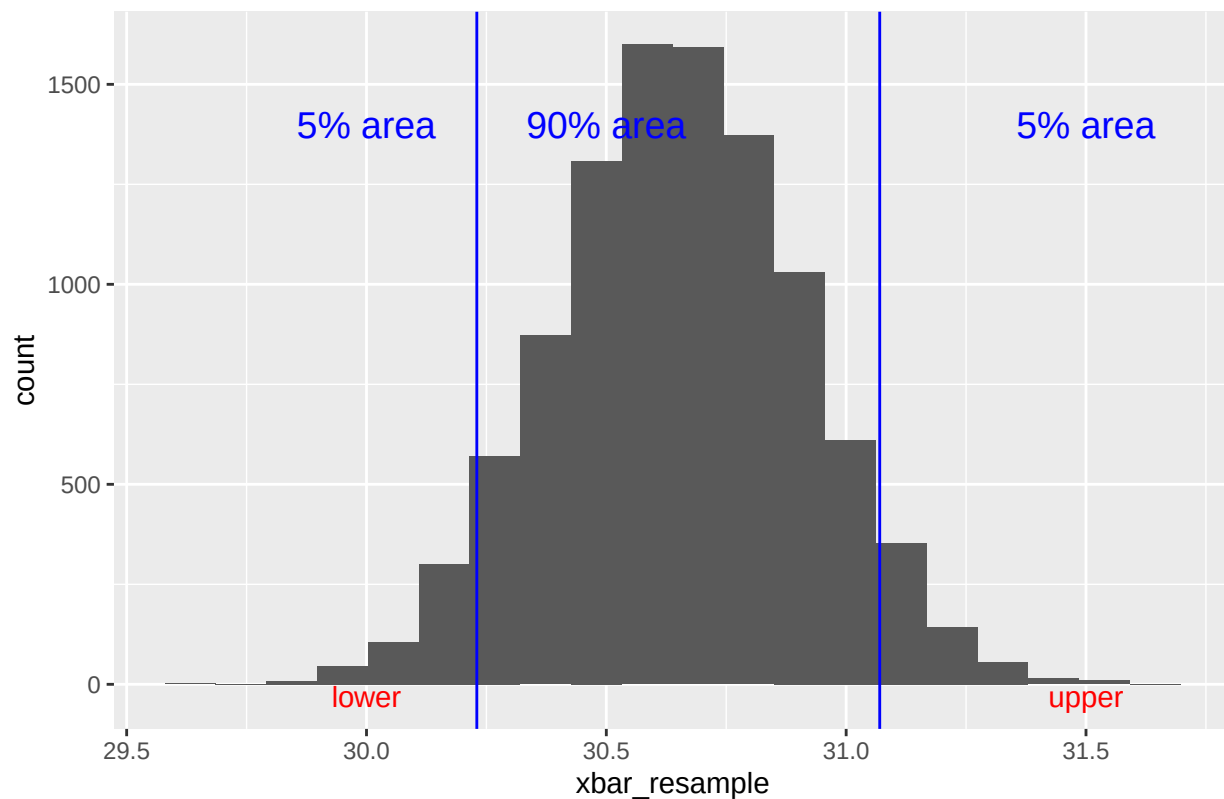


```
quantile(xbar_resample, c(0.05, 0.95))
```

```
##      5%      95%
## 30.22553 31.08082
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(30.23, 31.07), col = "blue")+
  ggtitle("90% Confidence interval of the population mean HEI score")+
  # Optional: Adds text on the plot
  annotate("text", x=30.5, y=1400, label= "90% area", color = 'blue', size =5)+
  annotate("text", x=30, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=31.5, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=31.5, y=-30, label= "upper", color = 'red', size =4)+
  annotate("text", x=30, y=-30, label= "lower", color = 'red', size =4)
```

90% Confidence interval of the population mean HEI score



The confidence interval calculated allows us to estimate the population mean HEI score

Specifically, we are 90% confident that the mean HEI score of customers in Minnesota is between 30.23 and 31.07

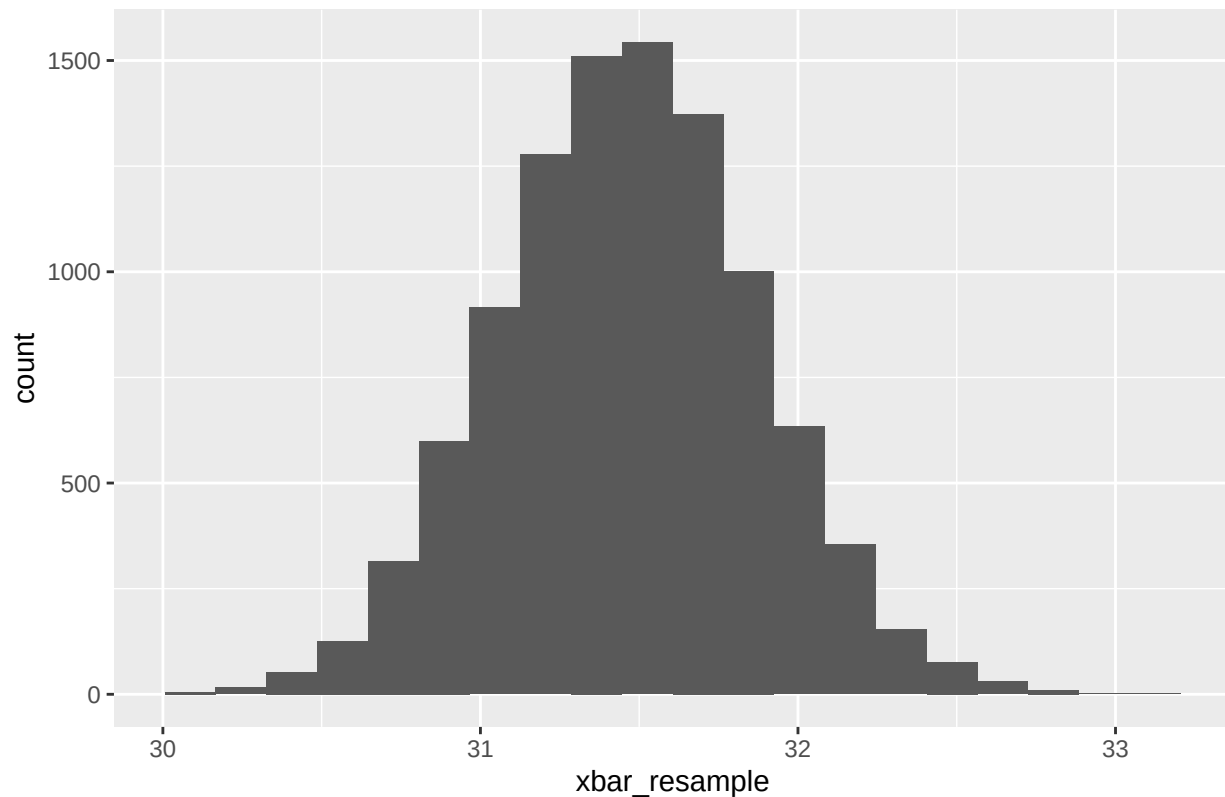
3. Estimate the mean HEI score (hei2010_total_score) of female customers in Minnesota. Bootstrap CI for a mean - Lecture 23.

```
customer_female = customer_intercep %>%
  filter(sex == "female")
```

```
xbar_resample <- replicate(10000, {
  customer_resample <- sample(customer_female$hei2010_total_score, replace = TRUE)
  mean(customer_resample)
})
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  ggtitle("Estimated sampling distribution of xbar")
```

Estimated sampling distribution of xbar

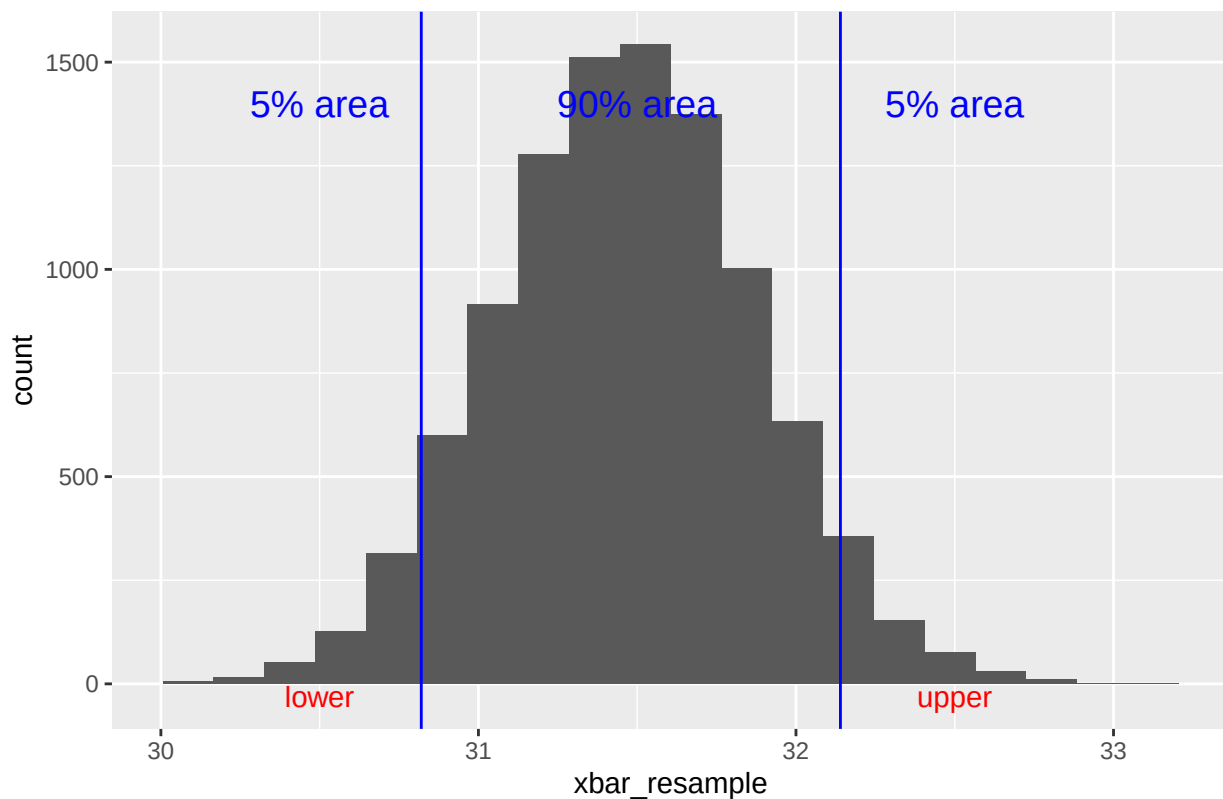


```
quantile(xbar_resample, c(0.05, 0.95))
```

```
##      5%      95%
## 30.79965 32.12941
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(30.82, 32.14), col = "blue")+
  ggtitle("90% Confidence interval of the female population mean HEI score")+
  # Optional: Adds text on the plot
  annotate("text", x=31.5, y=1400, label= "90% area", color = 'blue', size =5)+
  annotate("text", x=30.5, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=32.5, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=32.5, y=-30, label= "upper", color = 'red', size =4)+
  annotate("text", x=30.5, y=-30, label= "lower", color = 'red', size =4)
```

90% Confidence interval of the female population mean HEI score



The confidence interval calculated allows us to estimate the female population mean HEI score

Specifically, we are 90% confident that the mean HEI score of female customers in Minnesota is between 30.82 and 32.14, slightly higher than the total population.

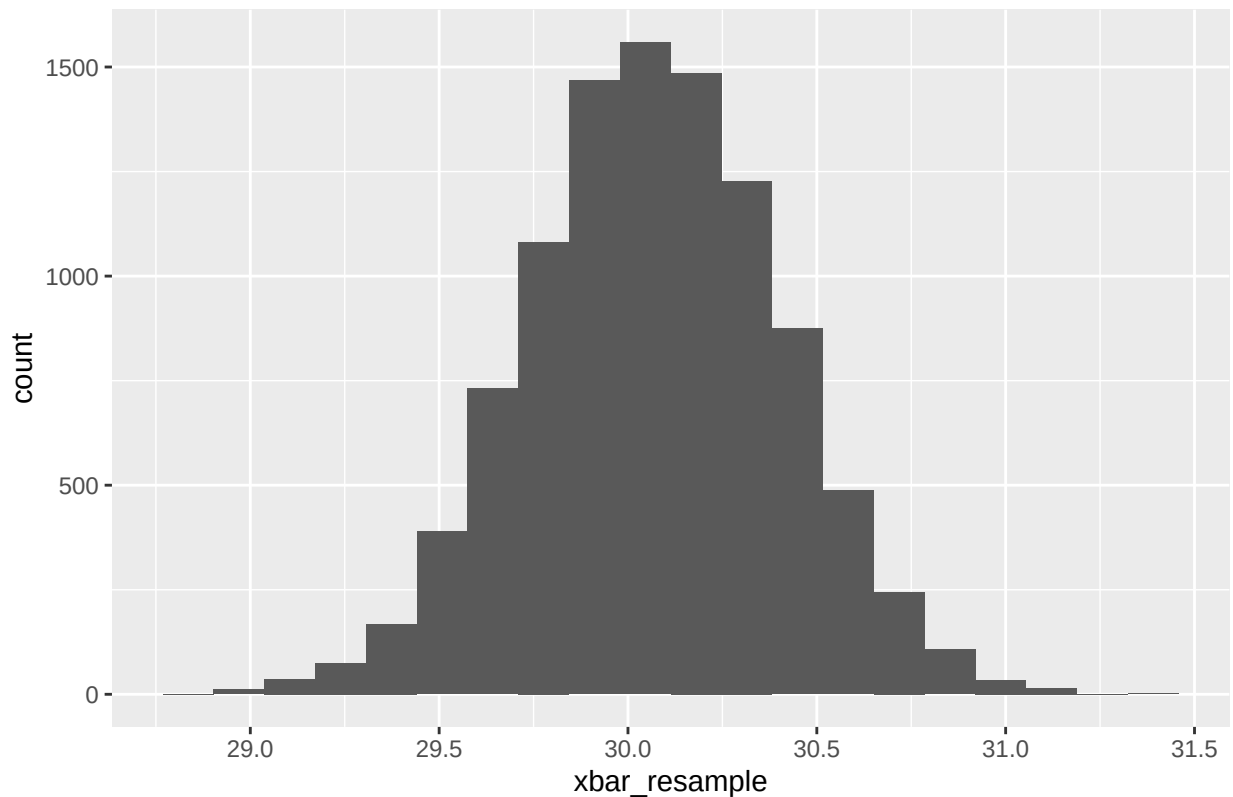
4. Estimate the mean HEI score (`hei2010_total_score`) of male customers in Minnesota. Bootstrap CI for a mean - Lecture 23.

```
customer_male = customer_intercep %>%
  filter(sex == "male")
```

```
xbar_resample <- replicate(10000, {
  customer_resample <- sample(customer_male$hei2010_total_score, replace = TRUE)
  mean(customer_resample)
})
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  ggtitle("Estimated sampling distribution of xbar")
```

Estimated sampling distribution of xbar

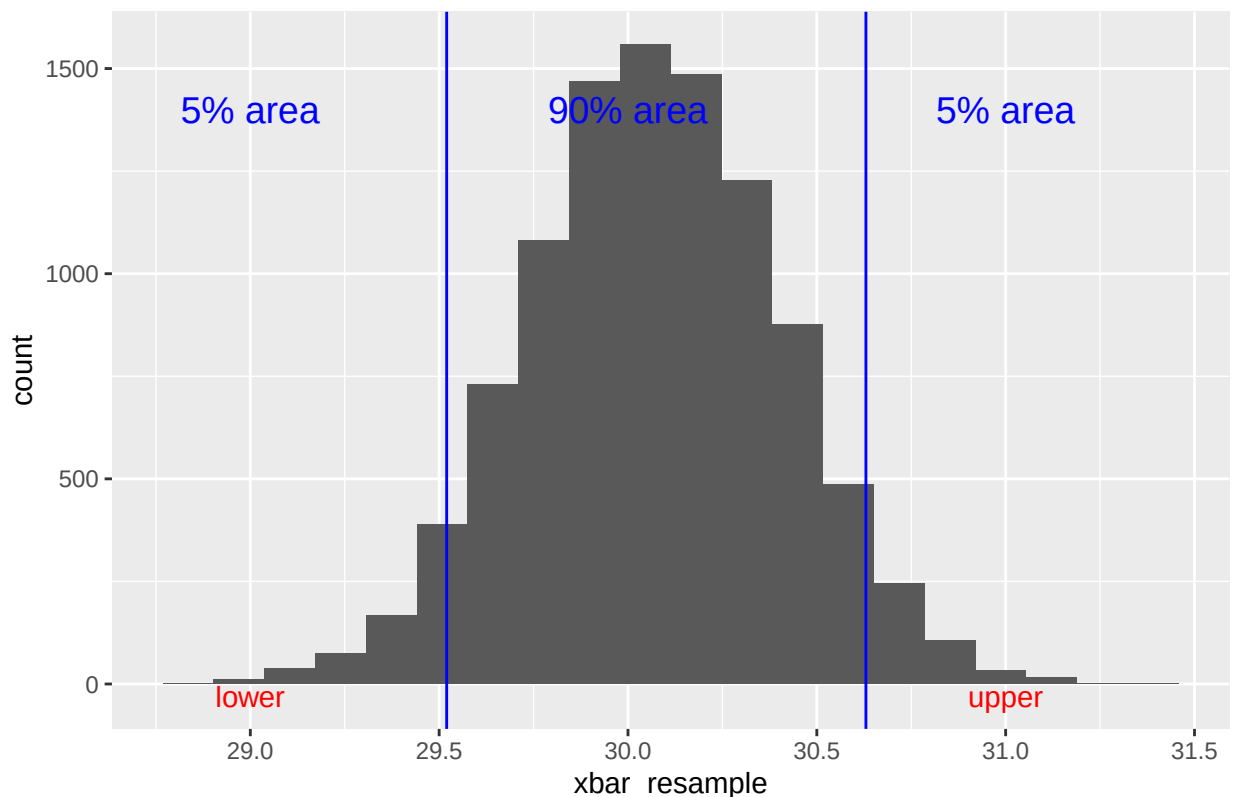


```
quantile(xbar_resample, c(0.05, 0.95))
```

```
##      5%      95%
## 29.52454 30.61800
```

```
ggplot(data.frame(xbar_resample), aes(x = xbar_resample)) +
  geom_histogram(bins=20) +
  geom_vline(xintercept = c(29.52, 30.63), col = "blue")+
  ggtitle("90% Confidence interval of the male population mean HEI score")+
  # Optional: Adds text on the plot
  annotate("text", x=30, y=1400, label= "90% area", color = 'blue', size =5)+
  annotate("text", x=29, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=31, y=1400, label= "5% area", color = 'blue', size =5)+
  annotate("text", x=31, y=-30, label= "upper", color = 'red', size =4)+
  annotate("text", x=29, y=-30, label= "lower", color = 'red', size =4)
```

90% Confidence interval of the male population mean HEI score



The confidence interval calculated allows us to estimate the female population mean HEI score

Specifically, we are 90% confident that the mean HEI score of male customers in Minnesota is between 29.52 and 30.63, slightly lower than the total population and lower than the female population.

5. Test if the mean HEI score of female customers is higher than for male customers. Permutation test for two means - Lecture 22.

Null hypothesis: There is no difference between the mean HEI scores for male and female customers. Alternative hypothesis: The mean HEI score for females is higher than the mean HEI score for males.

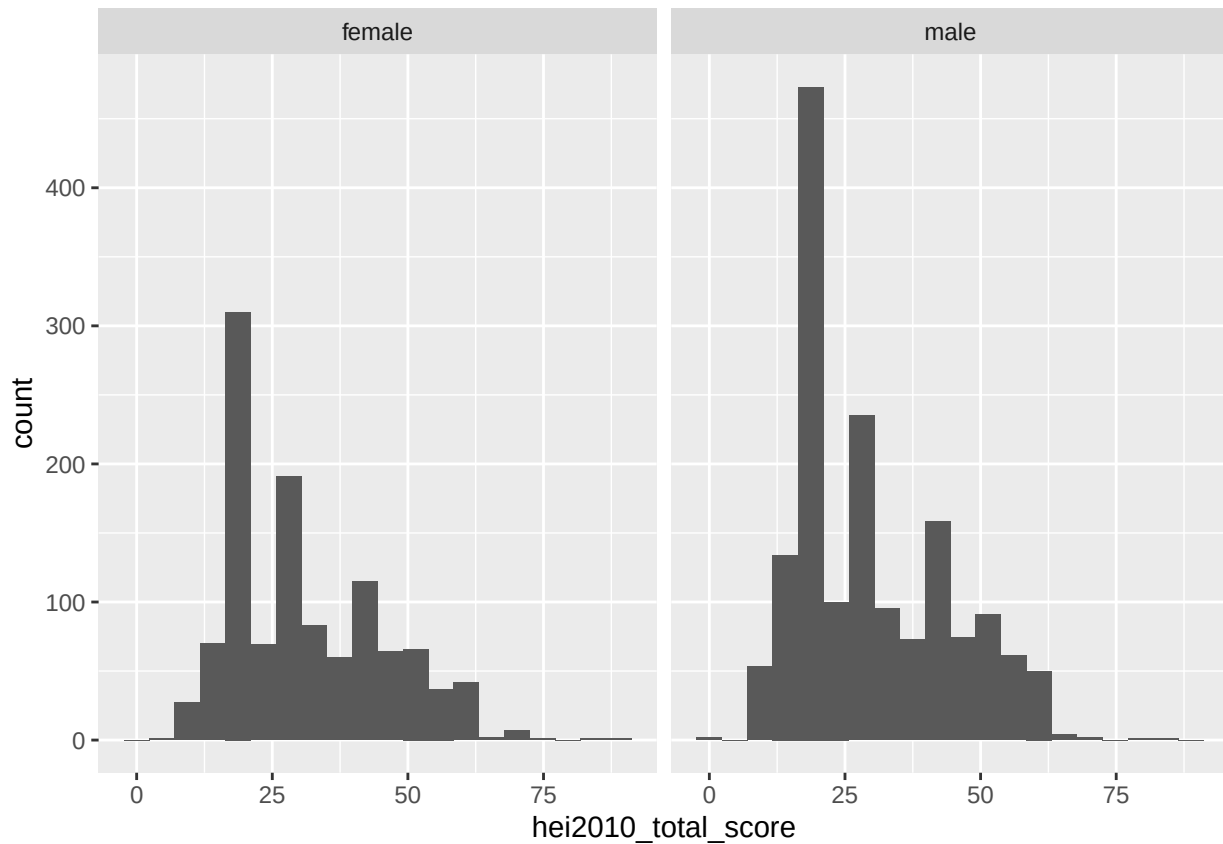
```
# A pipeline that finds the sample mean of group 1
xbar_1 <- customer_intercept %>%
  filter(sex == "female") %>%
  pull(hei2010_total_score) %>%
  mean()

# A pipeline that finds the sample mean of group 2
xbar_2 <- customer_intercept %>%
  filter(sex == "male") %>%
  pull(hei2010_total_score) %>%
  mean()

# The observed difference in sample means
xbar_diff <- xbar_1 - xbar_2
xbar_diff
```

```
## [1] 1.400839
```

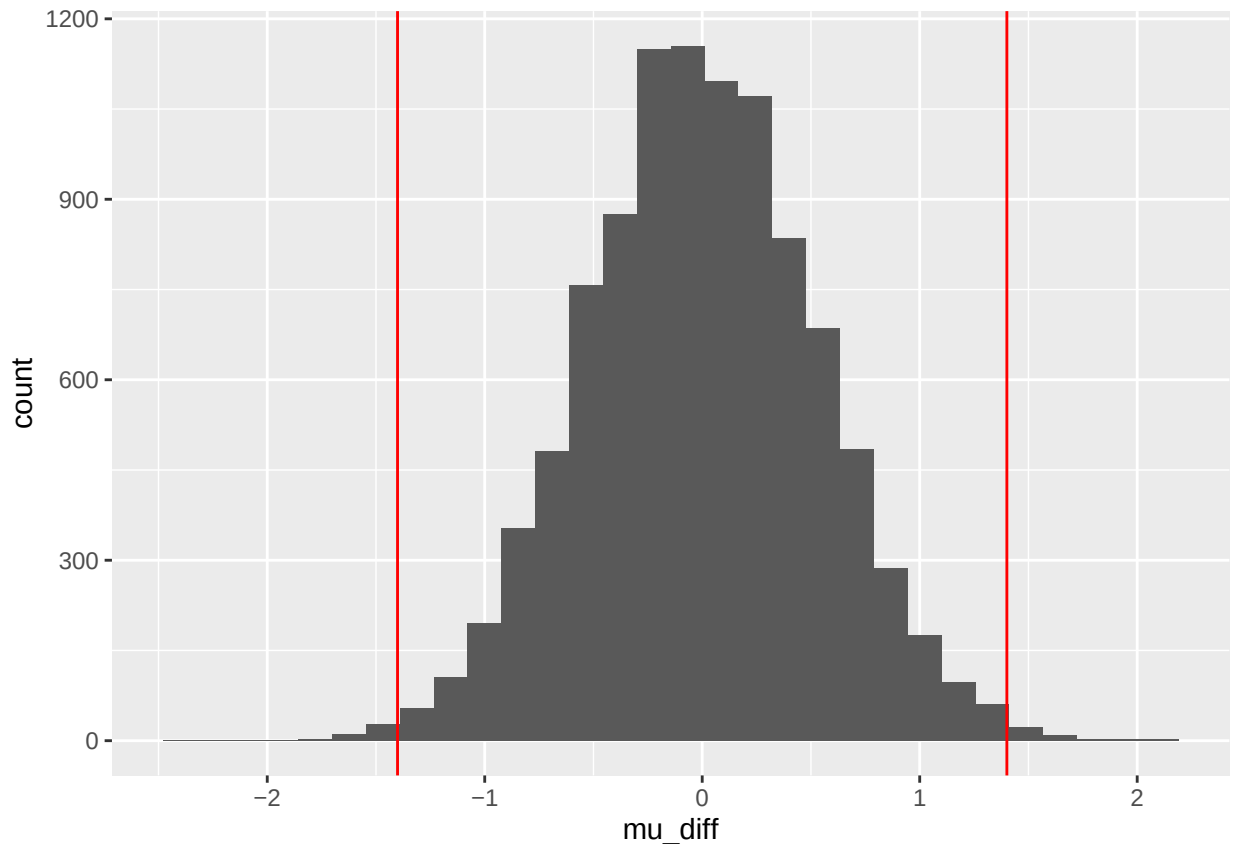
```
ggplot(customer_intercep, aes(x = hei2010_total_score)) +  
  geom_histogram( bins =20) +  
  facet_grid(~sex)
```



```
mu_diff <- replicate(10000, {  
  # generate new randomly permuted age groups  
  customer_intercep$permuted_sex <- sample(customer_intercep$sex, replace = FALSE)  
  
  # recompute the difference in means  
  mu_1 <- customer_intercep %>%  
    filter(permuted_sex == "female") %>%  
    pull(hei2010_total_score) %>%  
    mean()  
  mu_2 <- customer_intercep %>%  
    filter(permuted_sex == "male") %>%  
    pull(hei2010_total_score) %>%  
    mean()  
  mu_1 - mu_2  
})
```

```
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram() +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
pvalue <- mean(mu_diff > xbar_diff)
pvalue
```

```
## [1] 0.0045
```

The p value of 0.005 suggests that there is strong evidence to reject the null hypothesis and that the mean HEI score for females is indeed higher than that of males.

6. Test if the mean HEI score of College graduates is higher than for customers with a high school degree. Permutation test for two means - Lecture 22. Notice that here you will need to do the test on the subset of the data set that has only the education levels “Bachelors or higher” and “High school or less”

Null hypothesis: There is no difference in mean HEI scores between customers with a high school diploma and those with a bachelor’s degree or higher.

Alternative hypothesis: The mean HEI score for college grads is higher than that of those with only a diploma.

```
customer_education = customer_intercep %>%
  filter(education != "some college")
```

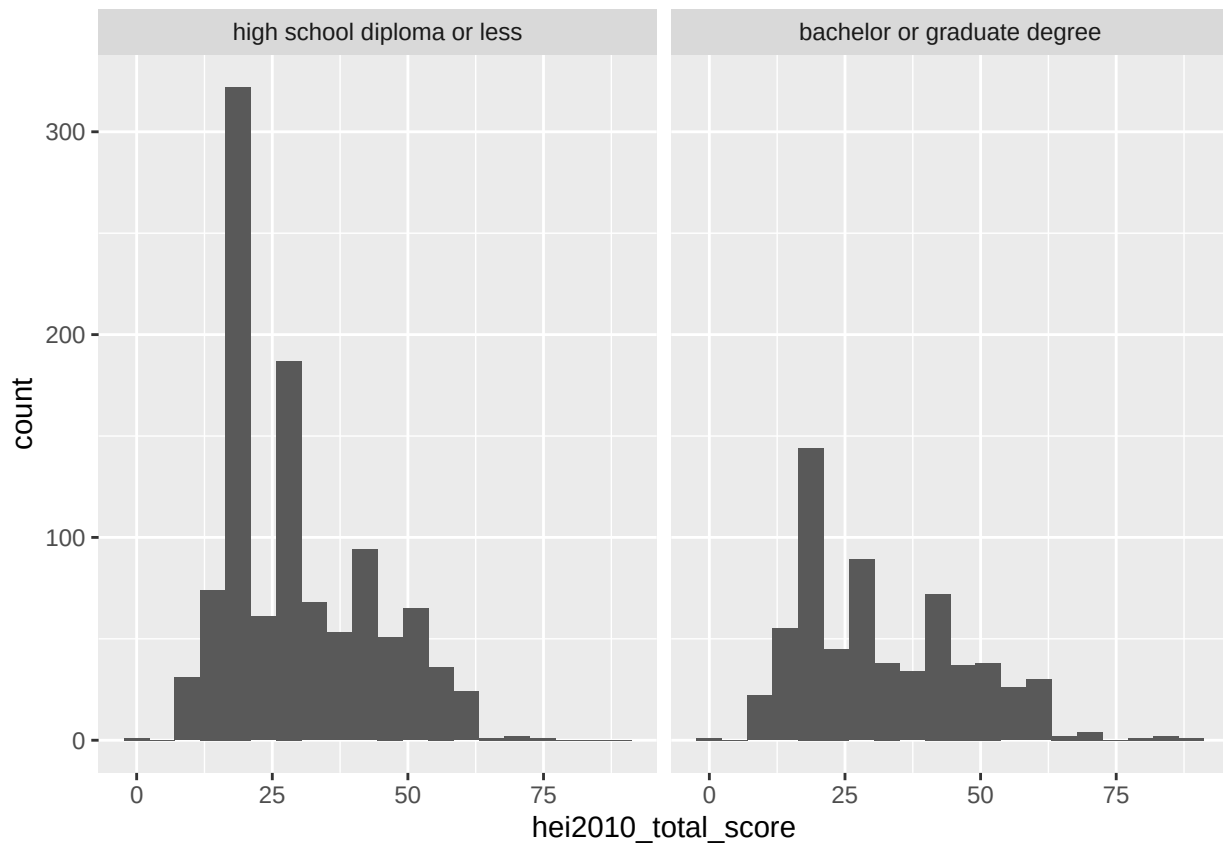
```
# A pipeline that finds the sample mean of group 1
xbar_1 <- customer_education %>%
  filter(education == "bachelor or graduate degree") %>%
  pull(hei2010_total_score) %>%
  mean()
```

```
# A pipeline that finds the sample mean of group 2
xbar_2 <- customer_education %>%
  filter(education == "high school diploma or less") %>%
  pull(hei2010_total_score) %>%
  mean()
```

```
# The observed difference in sample means
xbar_diff <- xbar_1 - xbar_2
xbar_diff
```

```
## [1] 2.218098
```

```
ggplot(customer_education, aes(x = hei2010_total_score)) +
  geom_histogram( bins =20) +
  facet_grid(~education)
```

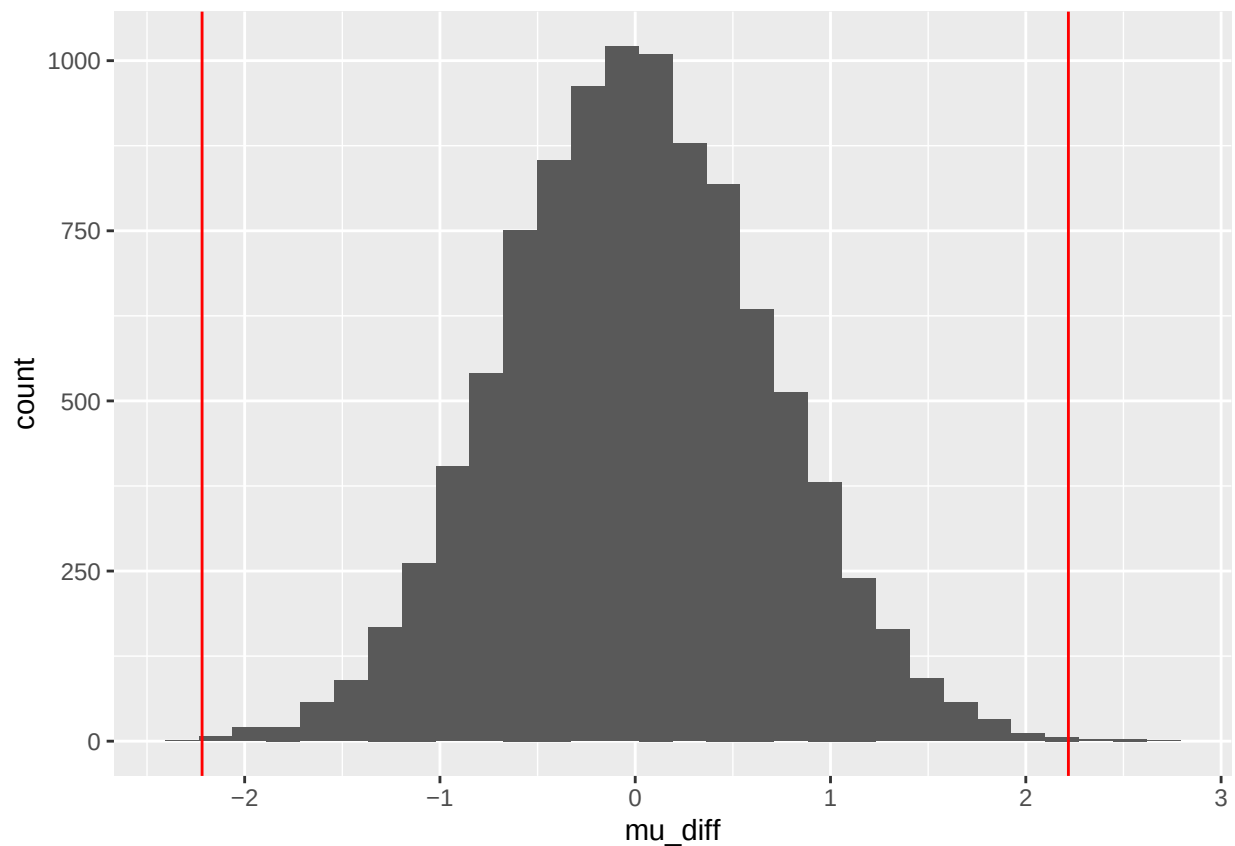


```
mu_diff <- replicate(10000, {
  # generate new randomly permuted age groups
  customer_education$permuted_edu <- sample(customer_education$education, replace = FALSE)

  # recompute the difference in means
  mu_1 <- customer_education %>%
    filter(permuted_edu == "bachelor or graduate degree") %>%
    pull(hei2010_total_score) %>%
    mean()
  mu_2 <- customer_education %>%
    filter(permuted_edu == "high school diploma or less") %>%
    pull(hei2010_total_score) %>%
    mean()
  mu_1 - mu_2
})
```

```
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram() +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.



```
pvalue <- mean(mu_diff > xbar_diff)
pvalue
```

```
## [1] 8e-04
```

The p value of 0.0006 suggests that we should reject the null hypothesis and that the mean HEI score of college grads is indeed higher than that of high school grads.

7. Test if the mean HEI score of customers that shop less than once a week is higher than customers that shop once per day. Permutation test for two means - Lecture 22. Again you will need to create an appropriate subset.

Null hypothesis: There is no difference in mean HEI scores between customers who shop less than once a week and those who shop once a day

Alternative hypothesis: The mean HEI score for customers who shop less than once a week is higher than that of those who shop every day.

```
customer_week = customer_intercep %>%
  filter(shopfreq3 != "1-6 times/wk")
```

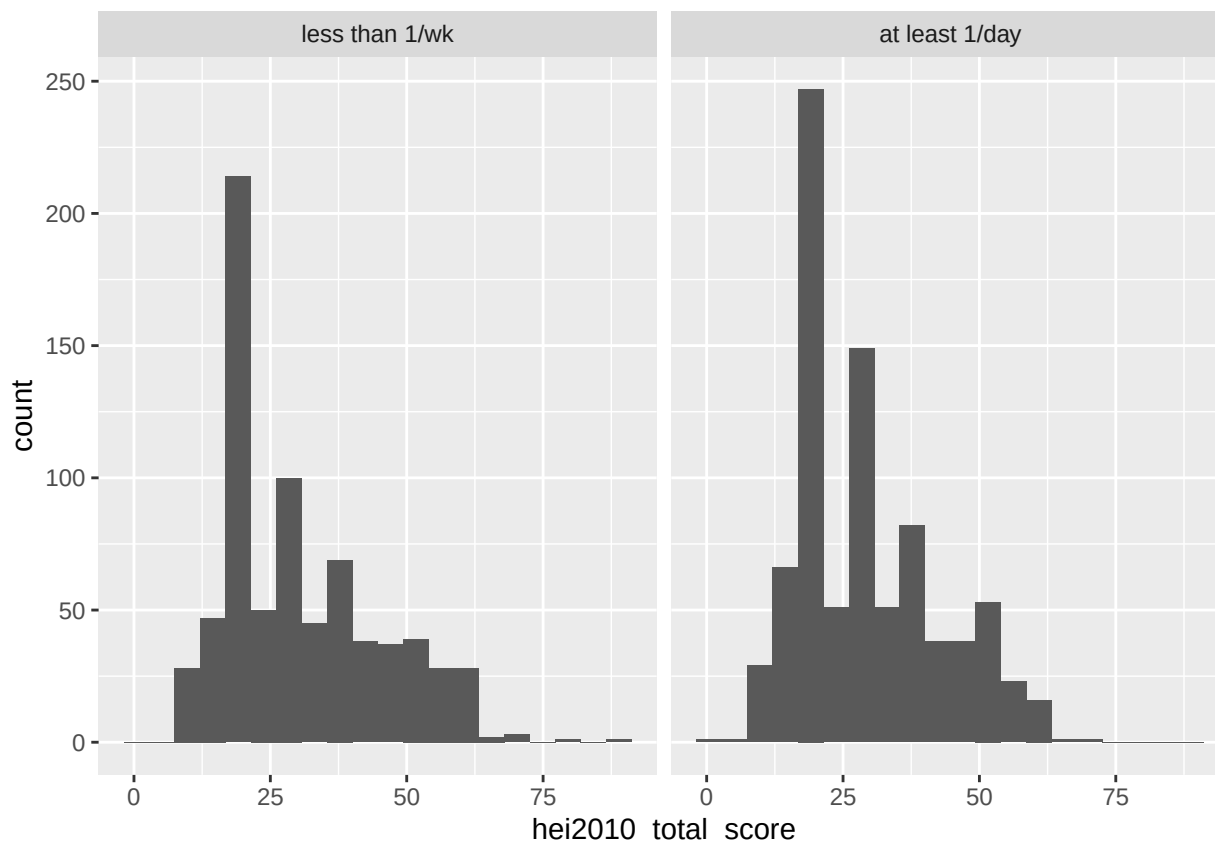
```
# A pipeline that finds the sample mean of group 1
xbar_1 <- customer_week %>%
  filter(shopfreq3 == "less than 1/wk") %>%
  pull(hei2010_total_score) %>%
  mean()
```

```
# A pipeline that finds the sample mean of group 2
xbar_2 <- customer_week %>%
  filter(shopfreq3 == "at least 1/day") %>%
  pull(hei2010_total_score) %>%
  mean()
```

```
# The observed difference in sample means
xbar_diff <- xbar_1 - xbar_2
xbar_diff
```

```
## [1] 1.281916
```

```
ggplot(customer_week, aes(x = hei2010_total_score)) +
  geom_histogram( bins =20) +
  facet_grid(~shopfreq3)
```

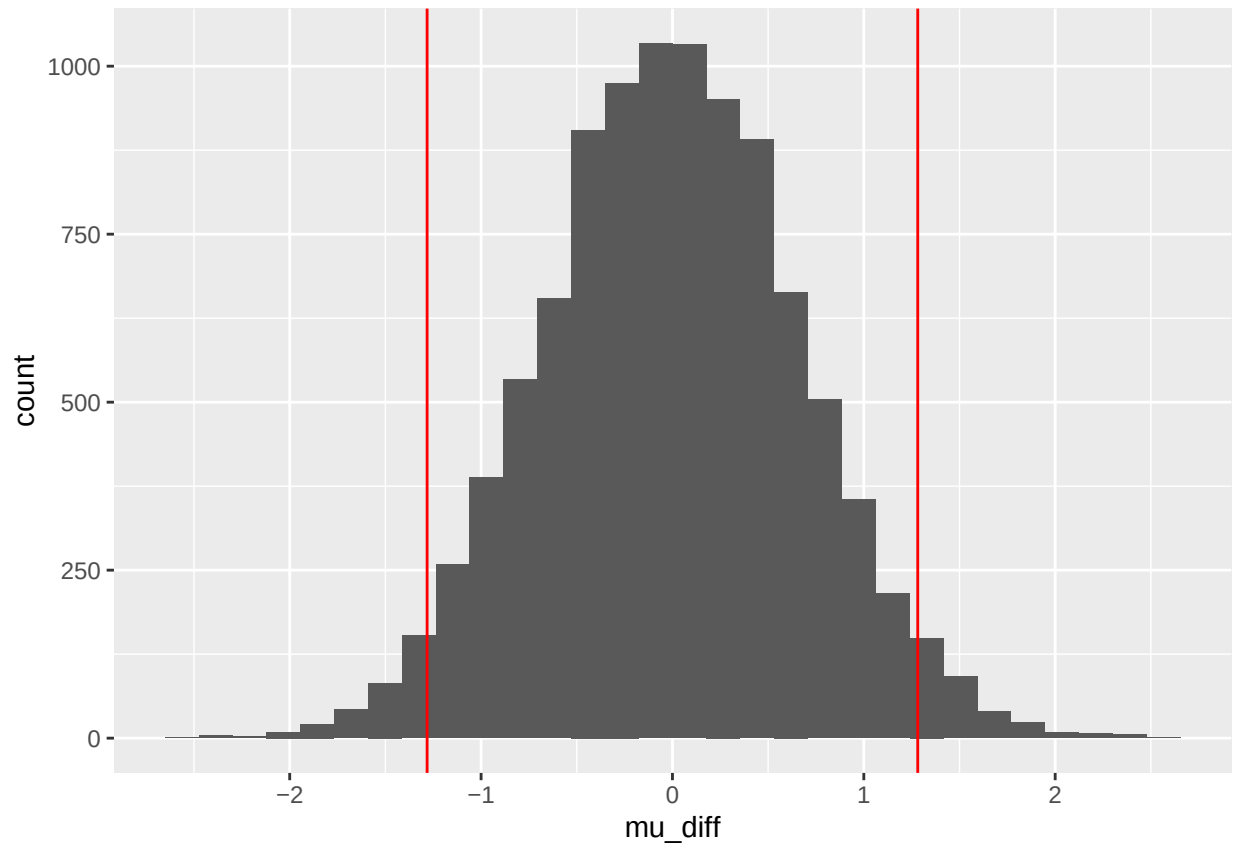


```
mu_diff <- replicate(10000, {
  # generate new randomly permuted age groups
  customer_week$permuted_week <- sample(customer_week$shopfreq3, replace = FALSE)

  # recompute the difference in means
  mu_1 <- customer_week %>%
    filter(permuted_week == "less than 1/wk") %>%
    pull(hei2010_total_score) %>%
    mean()
  mu_2 <- customer_week %>%
    filter(permuted_week == "at least 1/day") %>%
    pull(hei2010_total_score) %>%
    mean()
  mu_1 - mu_2
})
```

```
data.frame(mu_diff) %>% ggplot(aes(x = mu_diff)) +
  geom_histogram() +
  geom_vline(xintercept = xbar_diff, color = "red")+
  geom_vline(xintercept = -xbar_diff, color = "red")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
pvalue <- mean(mu_diff > xbar_diff)
pvalue
```

```
## [1] 0.0281
```

The p value of 0.027 suggests that we should reject the null hypothesis and that those who shop less than once a week have a higher mean HEI score than those who shop once a day.

8. (Optional but recommended) Apply the chisquare test of independence on the `hei_4groups` and `time` variable to test if the distribution of HEI score groups changed after the 1st time period.

Null hypothesis: The mean HEI score has not changed over time
 Alternative hypothesis: The mean HEI score has changed over time

```
table_hei_time = table(customer_intercep$hei_4groups, customer_intercep$time)
```

```
table_hei_time
```

```
##
##           1   2   3   4
## (-0.0888,22.2] 250 272 304 279
## (22.2,44.4]    260 294 297 293
## (44.4,66.6]    106 136 112 136
## (66.6,88.9]      0   3   9   3
```

```
chisq.test(table_hei_time)
```

```
## Warning in chisq.test(table_hei_time): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data:  table_hei_time
## X-squared = 15.509, df = 9, p-value = 0.07788
```

The p value of 0.077 suggests that we should fail to reject the null hypothesis, meaning it is likely that the mean HEI score has not changed over time.

9. What is your conclusion? Is the HEI score good for the Minnesota population? Which demographic groups are in danger? What are some published results about the HEI score on a national level. Is it important to provide more healthy options in small/medium food stores?

Based on the tests ran, the HEI score is not good for the Minnesota population. Women, those with college degrees, and those who shop only once a week had higher HEI scores, but they were still considered to be in the poor diet group. According to the USDA website, the mean HEI scores for the whole country was around 60, varying a bit depending on age. This suggests that there is a diet crisis in Minnesota and that the initiative to introduce more healthy options in stores was a well founded one.